



Traditional Chinese medicine toxicity prediction by heterogeneous network

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ABSTRACT

The clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity, and identifying toxic and effective herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity are complex and time-consuming. Given the large volume toxicity data of ingredients, building computational models for efficient ingredient-based toxicity prediction and evaluation is promising, but often fails to effectively predict the toxicity of herbs, due to the complexity herbs and toxic mechanisms. To address these challenges, we propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first constructs a heterogeneous network composed with herbs, ingredients and molecular targets, and leverages a heterogeneous graph attention network to learn the representation of herbs. Meanwhile, it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels, while pushing away the others. Next, it uses Multilayer Perceptron (MLP) on the herb representation to predict the toxic labels of this herb, and further introduces a weighted label fusion strategy that uses toxic labels of similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities, with 96% toxicity labels confirmed for canonical herbs. It can mine related toxic ingredients and targets in an interpretable way, and dissect the molecular mechanism of herb toxicity with authenticity.

1. Introduction

Herb medicine (Herb) has a long history in traditional medical practices around the world, especially in Asia, where it has been a cornerstone of healthcare for centuries (Hu et al., 2025). Herbs consist of a variety of bioactive ingredients, such as alkaloids, flavonoids, and terpenes, which contribute to their therapeutic effects. Although they are considered as safe and natural, many herbs have been linked to adverse effects, including hepatotoxicity, nephrotoxicity, cardiotoxicity, etc. (Jiang et al., 2010). These toxic effects can be caused by improper dosing, prolonged use, herb interactions, or contamination with heavy metals and pesticides. In addition, variations in herb quality arising from differences in cultivation, harvesting, and processing methods cause further complexity into their safety evaluation (Liu et al., 2020). Given these multifaceted challenges, there is an urgent need to develop effi-

cient and reliable methodologies to systematically identify and evaluate herb toxicity.

Different from the simple composition of chemical drugs, a herb contains a variety of bioactive ingredients with multi-target, multi-pathway and multi-effect interactions, and its toxicity mechanisms are more complex and diverse (Wu et al., 2021). Therefore, the study of herb toxicity faces some distinctive challenges. First, herb toxicity studies require complex and costly experiments. Data from different literature often exhibit substantial heterogeneity and inconsistency, while clinical data are scattered across institutions, lacking standardized formats and shared platforms. This makes it difficult to obtain high-quality toxicity data (Chan, 2003). Second, the multi-ingredient nature and complex toxicity mechanisms of herbs impose higher demands on models. However, traditional machine learning and deep learning methods struggle to process the multi-modal features of herbs (Wang et al., 2004). Third, herb

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toxicity prediction is often simplified as a binary classification task, whereas it should inherently be a multi-label classification task, as a herb is often associated with different toxic effects. In our collected TCM toxicity dataset, more than half of the herbs (141 out of 252) are linked with two or more toxic labels, highlighting the complexity and necessity of multi-label learning based herb toxicity prediction. Fourth, deep learning models are often seen as “black boxes”, making them difficult to provide clear biological explanations for their toxicity prediction (Gao et al., 2025). This lack of interpretability hinders researchers’ understanding of toxicity mechanisms, and thus undermines the authenticity and clinical applicability of deep models.

Regulatory authorities and researchers have been emphasizing the importance of improving the safety and quality control of herb products (Jitäreanu et al., 2022). Traditional methods for assessing herb toxicity, such as in vivo animal testing and in vitro cellular assays, are not only time-consuming and costly but also raise ethical concerns. Moreover, given the vast number of ingredients and their diverse action mechanisms, the comprehensive toxicity assessments remain a daunting task. With the advent of high-throughput technologies and the growing availability of large-scale biological and chemical datasets, computational approaches have gained attraction in herb toxicity prediction. These approaches include machine learning models, such as random forests, support vector machines, and deep learning frameworks, which can analyze complex, multidimensional data to predict toxicity outcomes (He et al., 2019a; Sun et al., 2019; Yang et al., 2021). By integrating data on chemical structures, pharmacological activities, and known toxicological profiles, these models offer a cost-effective and scalable solution for identifying potential toxicity of herbs and facilitating their safe usage (Jia et al., 2024; Wang et al., 2020; Wu et al., 2019a; Yu et al., 2025). However, existing computational approaches still encounter significant challenges arising from the heterogeneity and incompleteness of herb toxicity related data. Most current models are limited to a single type of information, such as molecular structure or pharmacological activity, and often lack the capability to effectively integrate diverse data sources that reflect the complex biological mechanisms of herbs. Furthermore, the absence of systematic frameworks capable of capturing multi-level interactions among herbs, ingredients, targets, and toxicities, restricts both the interpretability and authenticity of toxicity predictions. Given these, there is a critical need of authentic computational approaches that can systematically fuse multi-source information and address the challenges of data heterogeneity, sparsity, and interpretability in herb toxicity prediction.

In this paper, we construct a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies to solve the problem of dispersed and scarce herb toxicity data (see Section 4). To address the difficulties in modeling complex herbs and the lack of interpretability, we propose a heterogeneous network based approach (HerbToxNet) for herb toxicity prediction. HerbToxNet first constructs a herb-ingredient-target heterogeneous network using the pharmacological properties, toxicological characteristics, and associations between herbs, and then uses Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) to learn the representations of herbs. At the same time, it introduces contrastive learning with dynamic coefficient to pull closer the representations of herb pairs with shared toxic labels while pushing apart those with different toxic labels. Next, it uses a Multilayer Perceptron (MLP) on the herb representation to predict the initial toxic labels, and further introduces a weighted label fusion strategy that retrieves similar herbs and merges their toxic labels to refine the predicted labels.

Our contributions are summarized as below:

- (i) We construct and share a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies scattered in different herb databases and published literature.
- (ii) We propose a unified framework HerbToxNet to model multi-level relationships among herbs, ingredients, and targets via a heterogeneous network, which allows for transparent analysis of toxicity-related mechanisms and enhances interpretability.

- (iii) HerbToxNet synergizes heterogeneous network modeling, contrastive learning, and weighted label fusion to remedy scarce toxicity data and achieve toxicity prediction in a sensible and authentic way.
- (iv) Extensive experiments show that HerbToxNet outperforms competitive methods, identifies potential toxicities of canonical herbs (96% confirmed), and can directly uncover toxicity-related ingredients and targets from an interpretable view.

The rest of the paper is organized as follows. Section 2 discusses the related work and Section 3 elaborates on the proposed HerbToxNet framework. Section 4 presents the experimental setup, results and analysis. Section 5 summarizes our work and outlines future efforts.

2. Related work

Existing computational methods for herb toxicity usually predict some main types of toxicity, such as hepatotoxicity, nephrotoxicity, cardiotoxicity, etc., and model them as binary classification tasks (whether the interested toxicity is present or not). Based on the characteristics of herb toxicity, a series of theories and methods for herb toxicity prediction were developed, which can be divided into three categories: herb toxicity prediction based on quantitative structure-activity relationship (QSAR), network toxicology, herb property theory.

In computational toxicology, the virtual structure of a compound is often used to analyze, simulate, visualize, or predict the toxicity of a chemical by computational methods (Raies & Bajic, 2016). QSAR can quantitatively capture the relationship between molecular structure and biological effects, it empowers the computational prediction of compound toxicity and has been widely applied in the toxicity assessment of chemical drugs (Cherkasov et al., 2014). Therefore, QSAR-based methods can be adapted to computationally predict the toxicity of individual ingredients (alike drugs) within each herb, providing a basis for estimating the overall toxicity of herbs (Gao et al., 2019). He et al. (2019a) collected hepatotoxic herbs and ingredients, analyzed the structure of these ingredients and found that alkaloids and terpenoids were the main hepatotoxic ingredients. Yang et al. (2021) used QSAR to evaluate the potential toxicity of ingredients in Cassia seed, and clarified the relationship between structural characteristics and cytotoxicity by linear model, and found that three ingredients in Cassia seed had hepatotoxicity and eight ingredients had nephrotoxicity. Sun et al. (2019) used herb nephrotoxicity data to construct a QSAR model, and used ANN and SVM algorithms to predict nephrotoxicity. While QSAR-based approaches enable quantitative analysis of structure-toxicity relationships at the molecular level, their accuracy and applicability in herbal toxicity prediction are limited by the availability and quality of both structural and toxicological data, as well as the incomplete characterization of many ingredients.

Network toxicology, is an important branch for studying biomedicine based on network pharmacology (Li et al., 2019). The main research purpose of network toxicology is to analyze herb toxicity through a specific toxicity-target-ingredient-herb interaction network. The general analysis process includes: identification of toxic ingredients of herb, prediction of toxic ingredients and functional analysis of toxic targets. Wu et al. (2019a) combined four machine learning methods with four different fingerprint sets to construct 16 single classifiers to predict potential hepatotoxic ingredients, and explored the mechanism of hepatotoxicity of herbs through systematic pharmacological methods. Yu et al. (2025) compared ten machine learning models in the Zhengqing Fengtongning-induced liver injury dataset to find the most effective model for predicting liver injury. While network toxicology can systematically elucidate toxic mechanisms and identify key targets, this kind of methods cannot directly quantify herb-level toxicity and are limited by incomplete herb-ingredient-target associations, suffering biased or incomplete predictions.

The “Four Properties and Five Flavors” (also known as the “Four Qi and Five Tastes”) and meridian tropism constitute critical components of the theoretical framework of herb pharmacology (Qiao et al., 2022). The “Four Properties” of herb are categorized into five types: cold, hot,

warm, cool, and neutral. The “Five Flavors” are classified into seven types: sour, bitter, sweet, pungent, salty, bland, and astringent. Meridian tropism refers to classifying herbs into 12 channels: lung, large intestine, stomach, spleen, heart, small intestine, bladder, kidney, pericardium, triple burner, gallbladder, and liver. Wang et al. (2020) established a random walk network that can effectively interpret prescription information on the basis of traditional ANN to predict side effects of herb prescriptions. Jia et al. (2024) made toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. In addition, some studies have shown that the efficacy of herbs can be used to analyze herb toxicity (Yang et al., 2015; Zhu et al., 2025). While herb property theory offers descriptive insights rooted in traditional knowledge, its qualitative and subjective nature, distinct from the standardized representations used for chemical drugs, results in ambiguous data and limited predictive performance even when combined with advanced machine learning methods.

To address the above challenges, we construct a heterogeneous graph to model the complex relationships between herbs, ingredient and targets. By leveraging Heterogeneous Graph Attention Networks, contrastive learning with dynamic coefficient, and weighted label fusion, our HerbToxNet can handle incomplete data and fuse multi-level features, thereby effectively predicting the toxic labels of herbs.

3. The proposed methodology

The framework of HerbToxNet is illustrated in Fig. 1. HerbToxNet comprises four main modules: (i) Heterogeneous graph construction builds a herb-ingredient-target heterogeneous graph based on herb pharmacological properties, efficacy, associations; (ii) Herb representation learning employs a Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) with selected meta-paths to extract informative herb representations; (iii) Contrastive learning with dynamic coefficient optimizes the learned representations by pulling together herbs with shared

toxic labels and pushing apart those with dissimilar labels; and (iv) Toxicity prediction based on weighted label fusion merges the predictions from an MLP and toxic labels from similar herbs of the target herb based on their optimized representations. These modules are organized sequentially, enabling HerbToxNet to effectively integrate heterogeneous information, learn robust representations, and achieve toxicity prediction.

3.1. Heterogeneous graph construction

Herb pharmacological property and efficacy data contain useful information for toxicity prediction, but they are often incomplete or imprecise, as they are collected based on traditional literature and empirical knowledge. As such, existing methods built on herb pharmacological property and efficacy often struggle to exploit their potential. In fact, a herb has multiple ingredients with synergistic effects, making its toxicity complex and difficult to describe and capture with pharmacological property data alone (Caesar & Cech, 2019; Flora et al., 2013). Similarly, network toxicology faces challenges, as the complexity of ingredients makes it difficult for existing methods to accurately predict their toxic effects. Furthermore, these models often lack interpretability, hindering the understanding of biological mechanisms and reducing their reliability in practical applications (He et al., 2019a,b).

To address the above issues, we define the heterogeneous herb-ingredient-target graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which encode the multi-ingredient and multi-target nature of herbs by jointly modeling ingredient and target nodes, as well as their interactions. The node set $\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$ is the union of herb, ingredient, and target nodes, and the edge set $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ is the union of herb-ingredient, herb-target, ingredient-target, and herb-herb edges. We aim to integrate both the pharmacological properties and efficacy of each herb into its node representation, thereby combining domain knowledge with structural information of the heterogeneous graph. The pharmacological properties and efficacy of i th herb are denoted as $\mathbf{p}_i \in \{0, 1\}^{L_p}$ and $\mathbf{e}_i \in \{0, 1\}^{L_e}$,

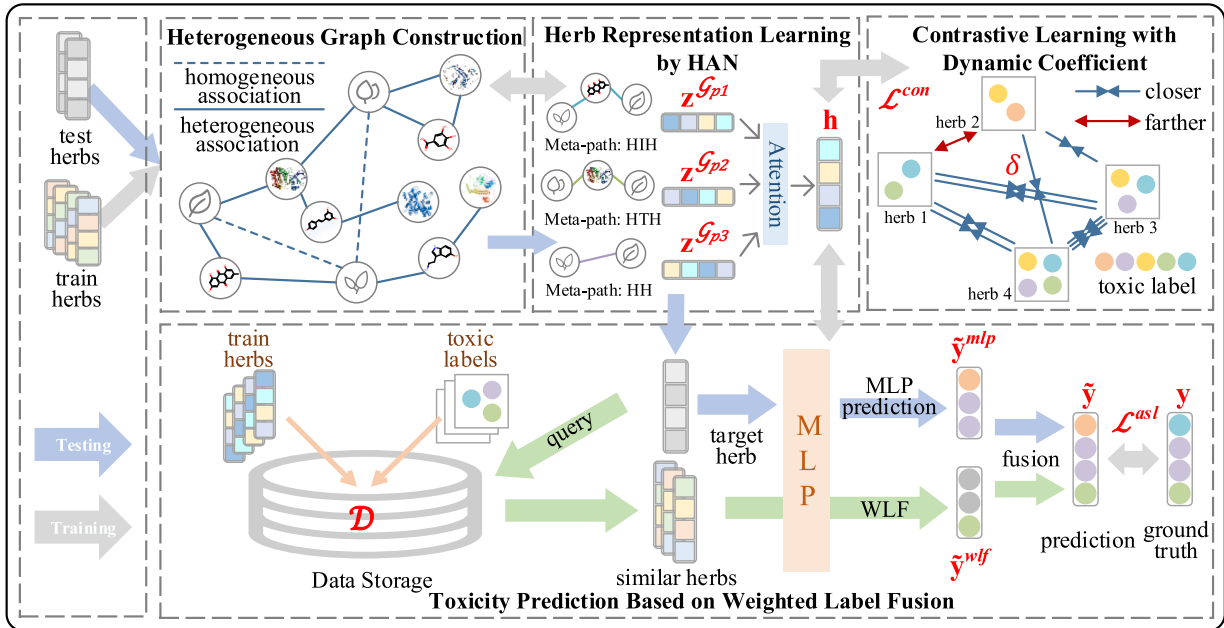


Fig. 1. The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph, and selects three types of meta-paths: herb-ingredient-herb (HIH), herb-target-herb (HTH), and herb-herb (HH) to build the meta-path matrix. Then, it utilizes the attention module of heterogeneous graph neural network to synthesize features ($\mathbf{z}^{Gp_1}$, $\mathbf{z}^{Gp_2}$, $\mathbf{z}^{Gp_3}$) under different meta-paths to obtain the representation ($\mathbf{h}$) of the herb. Next, it introduces a contrastive learning loss ($\mathcal{L}^{con}$) with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels, while pushing apart those without shared labels, thereby optimizing herb representations. After that, it predicts the toxic labels in two ways, an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$, and a weighted label fusion strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$, and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally, HerbToxNet defines asymmetric classification loss ($\mathcal{L}^{asl}$) between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

respectively, where L_p and L_e represent the dimension of pharmacological properties and efficacy. $p_{ij} = 1$ if i th herb possesses the j th pharmacological properties, and $p_{ij} = 0$ otherwise. $e_{ij} = 1$ if i th herb contains the j th efficacy, and $e_{ij} = 0$ otherwise. We use the pharmacological properties and efficacy of herbs as their initial node features. For the i th herb, its feature representation $\mathbf{f}_i$ is obtained by applying a transformation to the concatenation of $\mathbf{p}_i$ and $\mathbf{e}_i$ as:

$$\mathbf{f}_i = \text{ReLU}(\mathbf{W}_h(\mathbf{p}_i || \mathbf{e}_i) + \mathbf{b}_h) \quad (1)$$

where $\mathbf{W}_h$ and $\mathbf{b}_h$ denote the trainable weight matrix and bias vector, respectively. $||$ represents the concatenation operation, and $\text{ReLU}(\cdot)$ is the ReLU activation function. Since our objective is herb toxicity prediction, and the heterogeneous graph will be represented using HAN based on predefined meta-paths. Therefore, there is no need to assign specific initial features to the ingredient and target nodes.

The heterogeneous graph $\mathcal{G}$ consists of three node types: herb, ingredient, and target, which are connected by three types of heterogeneous relations (herb-ingredient, herb-target, and ingredient-target), as well as one type of homogeneous relation (herb-herb). Homogeneous relations between ingredients or targets require explicit similarity computations or manual curation, which introduce redundancy and tedious efforts, so these relations are not used in our work. This graph structure provides a unified tool for modeling both the structure and attribute information of herbs, facilitating toxicity prediction.

3.2. Herb representation learning by HAN

Give a meta-path p on the heterogeneous graph $\mathcal{G}$, with the meta-path taking the form of $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \dots \xrightarrow{r_l} v_{l+1}$, denoting that the node types v_1 and v_{l+1} are connected by a composite relationship $r = r_1 \circ r_2 \circ \dots \circ r_l$, where $\circ$ is the composite operator on the relationship, $v_i \in \mathcal{V}$ and $r_i \in \mathcal{E}$. In other words, the relationship r is composed of l relationships of those $r_1, r_2, \dots, r_l$. Thus, meta-path can capture specific semantic information within a heterogeneous graph, with different meta-paths representing different semantic contents.

For example, the meta-path of Herb-Ingredient-Herb (HIH) suggests that both herbs contain the same ingredient, implying that these two herbs may possess similar functionalities and target associations. We construct the matrix based on the HIH meta-path by multiplying the association matrix of herbs and ingredients in the heterogeneous graph with its transpose. Similarly, meta-path Herb-Target-Herb (HTH) suggests that both herbs link with the same target. The meta-path Herb-Herb (HH) indicates the similarity between two herbs. The construction methods for the HTH and HH matrices are similar to that of the HIH matrix.

Here, we decompose the heterogeneous graph $\mathcal{G}$ into multiple subgraphs, each subgraph $\mathcal{G}_p$ is composed of all neighbor pairs connected by the meta-path p , thereby designated as meta-path based subgraphs. Specifically, for the set of meta-paths that start or end with herb nodes, we denote it as $\mathcal{P} = \{p_1, p_2, \dots, p_M\}$, where $M = |\mathcal{P}|$ is the number of meta-paths considered.

The Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) is an effective tool for graph representation learning on heterogeneous graphs, capable of assigning different weights to different neighbors and meta-paths via a hierarchical attention mechanism. In this context, we utilize HAN to learn node representations within each meta-path induced subgraph. HAN adaptively aggregates the neighbor information of nodes in each subgraph to integrate both node topology and features. Specifically, for a given node and its associated subgraph, HAN first employs a node-level self-attention mechanism to learn the importance (attention scores) of its neighboring nodes. Then, it adaptively aggregates the feature representations of neighbors based on their attention scores. Finally, a semantic-level attention mechanism is used to combine representations from different meta-paths.

For the k th meta-path based subgraph $\mathcal{G}_{p_k}$, the importance $e_{ij}^{p_k}$ of node pair (i, j) can be formulated as follows:

$$e_{ij}^{p_k} = \text{LeakyReLU}(\mathbf{a}_{p_k}^T \cdot [\mathbf{f}_i^{p_k} || \mathbf{f}_j^{p_k}]) \quad (2)$$

where $\mathbf{a}_{p_k}$ is the attention parameter of the k th meta-path. To make the attention coefficients of different meta-paths comparable, we use the softmax function to normalize $e_{ij}^{p_k}$:

$$e_{ij}^{p_k'} = \frac{\exp(e_{ij}^{p_k})}{\sum_{j' \in \mathcal{V}_{hi}^{p_k}} \exp(e_{ij'}^{p_k})} \quad (3)$$

where $\mathcal{V}_{hi}^{p_k}$ represents the set of neighbor nodes of node i in the k th meta-path. Next, we aggregate the feature vectors of all meta-path instances using the ReLU activation function to derive the representation of node i based on the k th meta-path:

$$\mathbf{f}_i^{p_k} = \text{ReLU} \left(\sum_{j \in \mathcal{V}_{hi}^{p_k}} e_{ij}^{p_k'} \cdot \mathbf{f}_j^{p_k} \right) \quad (4)$$

In this study, we introduce a multi-head attention mechanism to enhance the stability of the attention coefficient learning process and mitigate the impact of individual attention heads. Specifically, we independently repeat the attention mechanism K times and concatenate the vector representations learned by each attention head. Thus, the representation $\mathbf{z}_i^{p_k}$ of node i under the k th meta-path is given by:

$$\mathbf{z}_i^{p_k} = \big\|_{j=1}^K \mathbf{f}_i^{p_k} \quad (5)$$

Generally, each node in the heterogeneous graph contains multiple types of semantic information, and the aggregation of nodes features under a single meta-path can only reflect one aspect of the node's information. To learn more comprehensive node embeddings, it is essential to integrate the semantic information represented by multiple meta-paths. A simple method for aggregating embeddings from different meta-paths is to take the average. However, due to the varying importance of meta-paths to node i in the heterogeneous graph, we adopt an attention mechanism to assign weights to each meta-path pattern and then perform aggregation. This ensures that the embeddings with the most crucial semantic information are given the most significance. More specifically, we learn the importance of each meta-path based subgraph and obtain $|\mathcal{P}|$ sets of node embeddings, where each set represents the weighted aggregation of all nodes under the corresponding meta-path:

$$w_{\mathcal{G}_{p_k}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh(\mathbf{W}_k \cdot \mathbf{z}_i^{p_k} + \mathbf{b}_k) \quad (6)$$

where $\mathbf{W}_k$, $\mathbf{b}_k$, and $\mathbf{q}_k$ denote the weight matrix, bias vector, and semantic-level attention vector associated with the k th meta-path. $\tanh(\cdot)$ represents the tanh activation function. After obtaining the attention scores of different meta-paths for a specific node, we normalize them using the softmax function, allowing for meaningful comparisons of the semantic information across different meta-paths.

$$w_{\mathcal{G}_{p_k}}' = \frac{\exp(w_{\mathcal{G}_{p_k}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{\mathcal{G}_{p_j}})} \quad (7)$$

Finally, we integrate the node embeddings from different meta-paths based on their respective attention scores. The i th herb node embedding $\mathbf{h}_i$ is as follows:

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w_{\mathcal{G}_{p_k}}' \cdot \mathbf{z}_i^{p_k} \quad (8)$$

3.3. Contrastive learning with dynamic coefficient

The above herb representations may not effectively capture toxicity-related similarities and differences among herbs, especially under setting where each herb can possess multiple toxic labels. Therefore, we propose a contrastive learning with dynamic coefficient strategy to update the herb representations. Canonical contrastive learning based methods aim to minimize the distance between instances of the same class while maximizing that between instances of different ones (Chen et al., 2020; Hadsell et al., 2006; Khosla et al., 2020). However, in multi-label herb toxicity prediction, two herbs often share only a subset of toxic labels, and each herb possesses unique toxic labels. This multi-label phenomenon imposes greater complexity than the single-label setting, making it hard to simply treat two herbs as either “positive” or “negative” pairs.

To capture the intricate relationships between herbs with multiple toxic labels, we define a dynamic coefficient based on toxic label semantic similarity. Unlike binary-valued indicators in single-label settings, this coefficient provides a flexible way to numerically quantify the degree of overlapped labels between herb pairs, allowing the model to account for the complex relationships inherent in multi-label toxicity prediction tasks. Specifically, we measure the feature similarity between two herbs as:

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\|_2 \|\mathbf{h}_j\|_2} \quad (9)$$

where $\cos(\dots)$ denotes the cosine similarity, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the learned embeddings of the i th and j th herbs, respectively, and $\|\cdot\|_2$ represents the L_2 norm.

To further incorporate toxic label relationships, we define the label semantic similarity β_{ij} as the fraction of shared toxic labels between two herbs:

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \quad (10)$$

where $\mathbf{y}_i \in \{0, 1\}^L$ and $\mathbf{y}_j \in \{0, 1\}^L$ are the multi-hot toxic label vectors for the i th and j th herbs, respectively, and L is the number of toxic labels. The numerator, $\mathbf{y}_i^T \cdot \mathbf{y}_j$, counts the number of shared toxic labels between the two herbs, while the denominator, $\mathbf{y}_i^T \cdot \mathbf{y}_i$, represents the total number of toxic labels for the i th herb. This ratio-based semantic similarity enables the model to distinguish not only completely identical or disjoint label sets in single-label setting, but also to quantify the partial similarity unique to multi-label scenario.

During subsequent experiments, we found that β may not sufficiently suitable to act as the dynamic coefficient for contrastive learning, as it does not provide a robust measure of relevance between two herbs. In particular, β can assign an excessively large coefficient to herb samples with low toxic label similarity. This issue can degrade the performance of herb toxicity prediction. As a result, we introduce a tunable hyperparameter ι to adjust β to obtain a more appropriate dynamic coefficient δ :

$$\delta_{ij} = (\beta_{ij})^\iota \quad (11)$$

The contrastive learning loss with dynamic coefficient δ_{ij} for a herb pair is defined as:

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\sum_{k=1, k \neq i}^N \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau)} \quad (12)$$

where τ is the temperature hyperparameter used to scale the similarity scores, and N is the total number of herbs in the batch. This formulation allows the model to emphasize similar herb pairs with higher toxicity similarity while effectively learning representations for herbs with multiple related toxic labels. This formulation explicitly leverages the multi-label toxicity of herbs, since each herb pair's contribution to the loss is scaled by the extent of their shared toxic labels.

For a given pair of herbs $(\mathbf{h}_i, \mathbf{h}_j)$, a higher semantic similarity β_{ij} results in a larger dynamic coefficient δ_{ij} , which in turn leads to a larger loss $\mathcal{L}_{ij}^{con}$. Consequently, the feature similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized toward a higher value. Conversely, if two samples do not share any labels ($\beta_{ij} = \delta_{ij} = 0$), the value $\mathcal{L}_{ij}^{con}$ becomes zero, and their cosine similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized to a lower value. The contrastive learning loss with dynamic coefficient for N herbs is computed as:

$$\mathcal{L}^{con} = \sum_{i=1}^N \sum_{j=1, j \neq i}^N \mathcal{L}_{ij}^{con} \quad (13)$$

3.4. Herb toxicity prediction based on weighted label fusion

For the i th herb, its representation $\mathbf{h}_i$ is passed through a three-layer MLP, followed by a sigmoid activation function, to compute the probability vector for all toxic labels:

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \quad (14)$$

where $\text{MLP}(\cdot)$ denotes a three-layer MLP. $\tilde{\mathbf{y}}_i^{mlp} \in \{0, 1\}^L$ represents the predicted probability vector that the i th herb is associated with toxic label.

However, the above MLP may not fully utilize the valuable knowledge encoded in the training data. To further enhance the toxicity prediction, we introduce a weighted label fusion (WLF) strategy. For each target herb, we retrieve training herbs with similar representations. By virtue of contrastive learning, these neighbors are more likely to share meaningful toxicity patterns with the target herb. HerbToxNet aggregates the toxic labels of neighborhood herbs in a weighted manner, which complements the MLP's output and contributes to more accurate toxicity prediction.

Let $\mathcal{D} = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ denote the training set with N herbs. Given the i th target herb with representation $\mathbf{h}_i$, we retrieve its relevant neighbors from the training set by computing the cosine similarity between $\mathbf{h}_i$ and all $\mathbf{h}_j$ in $\mathcal{D}$. Neighbors whose similarity exceeds a predefined threshold σ are selected to form the neighbor set:

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \quad (15)$$

Instead of using K-nearest neighbors, our threshold-based strategy adaptively selects neighbors that are close to the target herb in the representation space.

The WLF-based prediction for the i th herb is obtained by aggregating the label vectors of the selected neighbors, weighted by their similarities with respect to the target herb:

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \quad (16)$$

where the weight μ_j is defined as:

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \quad (17)$$

with τ' being a temperature hyperparameter that controls the sharpness of the weight distribution. In this way, neighbors with higher similarity to the i th herb are assigned with larger weights, which more significantly contribute to the WLF-based prediction than other neighbors.

To this end, we combine the MLP-based prediction $\tilde{\mathbf{y}}_i^{mlp}$ with WLF-based prediction $\tilde{\mathbf{y}}_i^{wlf}$ as:

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1 - \alpha) \tilde{\mathbf{y}}_i^{mlp} \quad (18)$$

where the scalar parameter α balances the two sources of predictions.

We adopt the asymmetric loss (ASL) (Ridnik et al., 2021) as the classification loss to address the imbalanced nature of multi-label herb toxicity prediction. ASL effectively mitigates the impact of label imbalance by assigning different penalties to positive and negative samples, thereby enhancing the model's focus on rare but critical toxic labels. The loss

function is defined as follows:

$$\mathcal{L}^{asl} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L \left[y_{ij} (1 - \tilde{y}_{ij})^{\gamma^+} \log(\tilde{y}_{ij}) + (1 - y_{ij}) (p_{ij}^m)^{\gamma^-} \log(1 - p_{ij}^m) \right] \quad (19)$$

$$p_{ij}^m = \max(\tilde{y}_{ij} - m, 0) \quad (20)$$

where γ^+ and γ^- are focusing parameters for positive and negative samples, respectively, and $m \geq 0$ is the probability margin.

The overall loss function of HerbToxNet consists of the ASL classification loss and the contrastive learning loss, formulated as:

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda \mathcal{L}^{con} \quad (21)$$

where λ is a the hyperparameter that balances the two losses. Algorithm 1 summarizes the pseudo code of HerbToxNet

Algorithm 1 The overall process of HerbToxNet.

Input: herb pharmacology $\mathbf{p}$, efficacy $\mathbf{e}$; toxic labels $\mathbf{y}$; meta-path set $\mathcal{P}$; node sets $\mathcal{V}$; edge sets $\mathcal{E}$; model parameters Θ ;

Output: Predicted herb toxic label $\tilde{\mathbf{y}}$;

- 1: $\triangleright$ Heterogeneous Graph Construction
 - 2: Build node set $\mathcal{V}_h$ (herb), $\mathcal{V}_c$ (ingredient), $\mathcal{V}_t$ (target) and edge set $\mathcal{E}_{hc}, \mathcal{E}_{ht}, \mathcal{E}_{ct}, \mathcal{E}_{hh}$;
 - 3: For each herb i , initialize node features $\mathbf{f}_i$ via Eq. (1);
 - 4: Assemble the heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$;
 - 5: $\triangleright$ Herb Representation Learning by HAN
 - 6: **for** each meta-path $p_k \in \mathcal{P}$ **do**
 - 7: Extract meta-path subgraph $\mathcal{G}_{p_k}$ from $\mathcal{G}$;
 - 8: **for** each node i **do**
 - 9: Compute attention score $e_{ij}^{p_k}$ via Eq. (2) and normalize it via Eq. (3);
 - 10: Get neighbor-aggregated features $\mathbf{f}_i^{p_k}$ via Eq. (4);
 - 11: Obtain multi-head feature representation $\mathbf{z}_i^{p_k}$ via Eq. (5);
 - 12: **end for**
 - 13: **end for**
 - 14: Compute meta-path attention w_{p_k} via Eq. (6) and normalize it via Eq. (7);
 - 15: Fuse meta-path embeddings to get $\mathbf{h}_i$ via Eq. (8);
 - 16: $\triangleright$ Contrastive Learning with Dynamic Coefficient
 - 17: **for** each herb pair (i, j) **do**
 - 18: Compute feature similarity via Eq. (9);
 - 19: Compute semantic similarity β_{ij} and dynamic coefficient δ_{ij} via Eq. (10) and Eq. (11);
 - 20: **end for**
 - 21: Compute total contrastive loss $\mathcal{L}^{con}$ via Eq. (13);
 - 22: $\triangleright$ Herb Toxicity Prediction
 - 23: **for** each herb i **do**
 - 24: MLP predicts $\tilde{\mathbf{y}}_i^{mlp}$ via Eq. (14);
 - 25: Weighted fuse $\tilde{\mathbf{y}}_i^{mlp}$ and $\tilde{\mathbf{y}}_i^{wlf}$ into $\tilde{\mathbf{y}}_i$ via Eq. (18);
 - 26: **end for**
 - 27: Compute asymmetric loss $\mathcal{L}^{asl}$ via Eq. (19) and Eq. (20);
 - 28: Compute total loss $\mathcal{L}$ via Eq. (21);
 - 29: Update model parameters Θ by back propagation and Adam;
 - 30: return Predicted toxic labels of herb $\tilde{\mathbf{y}}$;
-

4. Experiment

4.1. Experimental setup

4.1.1. Dataset

We collected pharmacological properties, efficacy, and interaction data for herbs from established databases, including ChP (2020)

(Xu et al., 2021), HERB (Fang et al., 2021), SymMap v2 (Wu et al., 2019b), TCMBank (Lv et al., 2023), and HIT 2.0 (Yan et al., 2022b). Toxicity labels were obtained from major literature and database sources such as Google Scholar, CNKI, VIP, WanFang, PubMed, and Web of Science, and were grouped into five categories: cardiotoxicity, nephrotoxicity, hepatotoxicity, systemic toxicity, and other toxicity. Table 1 gives an overview of the data sources.

The curation pipeline of our dataset is reported in Section I of Supplementary file. Then, we take known toxic labels of a herb as positive samples and others as negative ones of this herb. We keep 15% of the herbs as the testing set, while the remaining herbs are used for five-fold cross-validation. This process is independently repeated ten times to ensure robustness. To ensure the integrity, all interactions associated with a herb are grouped into the same subset during the partition process.

4.1.2. Evaluation metrics

To evaluate the performance of HerbToxNet and the baseline methods, we employed four widely used evaluation metrics (Qiu et al., 2023): Macro-averaged F1 Score (Macro-F1), Micro-averaged F1 Score (Micro-F1), Average Area Under the receiver operating characteristic Curve (AvgAUC), and One Error (OneError). Macro-F1 independently calculates the F1 score for each label and then takes the average, reflecting the overall performance across all labels. Micro-F1 aggregates the contributions of all labels to compute the average F1 score, reflecting overall performance on all samples. AvgAUC computes the AUC for each label and then takes the average, a higher value means a better separability between toxic and non-toxic labels. OneError counts the number of the predicted toxic label with top-rank is not among the true labels; a lower values indicate that the model is more likely to rank a true label first. For consistency with other metrics, we use $1 - \text{OneError}$ instead.

4.1.3. Baseline methods

To prove the effectiveness of our proposed HerbToxNet, several representative methods were selected as the baseline, which can be summarized into three categories: herb toxicity prediction methods, chemical drug toxicity prediction methods, and multi-label prediction methods in the field of biomedicine. (i) Herb toxicity prediction methods: ZF-LightGBM (Yu et al., 2025) compares ten machine learning models on *Zhengqing Fengtongning*-induced liver injury dataset and finds LightGBM performing the best; QSAR-ANN and QSAR-SVM (Sun et al., 2019) uses herb nephrotoxicity data to construct QSAR model, and apply ANN and SVM algorithms to predict nephrotoxicity, respectively; EI-MS-RF (Jia et al., 2024) makes toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. (ii) Chemical drug toxicity prediction methods: DeepDILI (Li et al., 2020) combines model-level representations generated by traditional machine learning algorithms with a deep learning framework based on Mold2 descriptors to predict drug-induced liver injury; R-E-GA (Yan et al., 2022a) rotates the fusion vector represented by a high-dimensional molecule in the feature space to find a subset of features with better predictive performance, and then integrates an Adaboost-based ensemble learning method into R-E-GA to improve prediction accuracy; TIRPnet (Wei et al., 2024) is designed to decipher knowledge from abundant of TCM spontaneous reports, enabling the prediction of potential adverse drug reactions (ADRs) for individual ingredient. (iii) multi-label prediction methods: ML-PRDF (Gong et al., 2023) initially utilizes PCC-MLRF (Gong et al., 2023) for feature selection, and then inputs the meticulously selected features into the multi-label deep forest model to conduct differentiation and predictive analysis; MulGraL (Pham et al., 2022) constructs a heterogeneous information network relevant to the prediction outcomes, subsequently combines multiple multi-label classifiers to enhance the accuracy and reliability of prediction; MLC-CNN (Ge et al., 2021) involves grouping multi-labels according to their inter-label correlations and utilizes neural network models to perform predictions for

Table 1
Details of the collected data sources.

	Type	Size	Sources
Herb	Pharmacological Property	252 × 33	Chp (2020), HERB, SymMap v2, TCM-Bank, HIT 2.0
	Efficacy	252 × 331	
	Herb-Target	34988	
	Herb-Ingredient	3275	
	Ingredient-Target	5932	
Toxicity	Cardiotoxicity	74	Google Scholar, CNKI, VIP, WanFang, PubMed, WOS
	Nephrotoxicity	101	
	Hepatotoxicity	114	
	Systemic Toxicity	167	
	Other Toxicity	150	

Note: The size of 'Efficacy' is 252 × 331, indicating that the number of herbs is 252 and the number of efficacy is 331. 'Pharmacological Property' is represented in the same way as 'Efficacy'.

each group, effectively leveraging the impact of diverse features on distinct labels.

Since the first two categories of methods were originally developed for binary classification tasks, we modify their final binary outputs to produce multi-label predictions suitable for our task. In addition, as EI-MS data are not available in our dataset, we substitute the EI-MS descriptors with herb pharmacological property and efficacy data, and implement the same EI-MS-RF pipeline as described in the original work. For each baseline method, we used the recommended parameters as a starting point and fine-tuned them as necessary for our dataset. Detailed parameters for all baselines and our HerbToxNet, as well as a comprehensive parameter sensitivity analysis of HerbToxNet, are presented in Section III of Supplementary file.

4.2. Results comparison with existing solutions

Table 2 summarizes the performance of HerbToxNet and baselines. HerbToxNet achieves the best herb toxicity prediction performance in most evaluation metrics. Other important observations are:

(i) HerbToxNet significantly outperforms other herb toxicity prediction methods, including ZF-LightGBM, QSAR-SVM, EI-MS-RF, and QSAR-ANN, primarily due to their inability to effectively capture the complex multi-ingredient and multi-target characteristics of herbs. This defect limits their generalization ability, which is particularly evident when handling multi-label toxicity prediction tasks. Moreover, they typically predict single toxic labels and fail to effectively model the intricate correlations between toxic labels, thereby restricting their performance in multi-label toxicity prediction. In contrast, HerbToxNet achieves better performance in complex multi-label toxicity prediction tasks, significantly leading in most evaluation metrics. Its advantages are as follows. First, HerbToxNet constructs a herb-ingredient-target heterogeneous network and utilizes meta-path to extract properties and rela-

tional information of herbs, achieving precise modeling of complex characteristics of herbs. Second, it introduces a contrastive learning with dynamic coefficient strategy that brings closer the representations of herb pairs with similar toxic labels while pushing apart those with different toxic labels, enhancing the model's adaptability to multi-label tasks. Third, HerbToxNet introduces a weighted label fusion strategy to effectively utilize the toxic labels of similar herbs, thus further improving the prediction performance and alleviating the impact of scarce herb data. (ii) HerbToxNet also significantly outperforms drug toxicity prediction methods, including DeepDILI, R-E-GA, and TIRPnet. These methods perform poorly in the multi-label toxicity prediction task for herb, because chemical drugs typically possess well-defined molecular structures and fingerprints, which provide abundant and detailed information for prediction. In contrast, a herb comprises of multi-ingredients with lower-dimensional pharmacological data, and the one-hot encoding to represent these ingredients is insufficient to capture their complex characteristics. Moreover, they may not effectively capture and leverage the correlations between toxic labels associated with herbs. HerbToxNet addresses these challenges by leveraging heterogeneous network modeling, contrastive learning with dynamic coefficient, and weighted label fusion. These innovations enable HerbToxNet to effectively capture the intricate features of herbs and significantly enhance its performance in multi-label toxicity prediction tasks.

(iii) Multi-label classification-based methods show certain advantages in predicting the toxic labels of herbs, compared with the other two kinds of methods. That is because they can consider multiple toxicity labels simultaneously, which is naturally suited for multi-label herb toxicity prediction. Although these methods can provide competitive results in some scenarios, they may be limited in mining the intrinsic information and complex characteristics of limited herb data. Therefore, as shown in Table 2, these methods again lose to HerbToxNet. The superior performance of HerbToxNet is mainly attributed to its effective modeling

Table 2
Performance comparison of HerbToxNet and baseline methods in predicting herb toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.2371±0.0413	0.4127±0.0420	0.5163±0.0307	0.2535±0.0487
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536
QSAR-SVM	0.1619±0.0235	0.3842±0.0422	0.5132±0.0422	0.2582±0.0520
EI-MS-RF	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.0482
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.0702
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417
TIRPnet	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608
ML-PRDF	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611
MLC-CNN	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676
	0.5652±0.0161	0.6247±0.0172	0.6003±0.0141	0.5587±0.0485

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

Table 3

Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.

	Hepatotoxicity Prediction			Nephrotoxicity Prediction		
	AUROC	AUPRC	F1	AUROC	AUPRC	F1
ZF-LightGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
QSAR-SVM	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
EI-MS-RF	0.5624±0.0765	0.5304±0.0984	0.6435±0.0594	0.5431±0.0743	0.4520±0.0782	0.6023±0.0566
DeepDILI	0.5486±0.0763	0.5658±0.1005	0.6272±0.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
R-E-GA	0.5342±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
TIRPnet	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
HerbToxNet	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

of herbs, ingredients and targets by heterogeneous network, contrastive learning and weighted label fusion.

4.3. Hepatotoxicity and nephrotoxicity prediction

HerbToxNet is designed for multi-label toxicity prediction, while most methods target at single-label toxicity prediction of herbs (Jia et al., 2024; Sun et al., 2019; Yu et al., 2025) and of drugs (Li et al., 2020; Wei et al., 2024; Yan et al., 2022a), they build on binary tasks for single type toxicity. To assess whether HerbToxNet can be adapted to these binary tasks, we isolated hepatotoxicity and nephrotoxicity herbs from the dataset, divided them as hepatotoxicity and nephrotoxicity datasets and evaluated our approach in both datasets. Among them, the hepatotoxicity dataset contains 114 herbs and the nephrotoxicity dataset contains 101 herbs. Since this is a binary task, AUROC, AUPRC, and F1 are used for evaluation.

As shown in Table 3, although HerbToxNet is originally designed for multi-label toxicity prediction, it demonstrates good adaptability and robustness when applied to binary classification tasks focused on individual toxicity types. In the hepatotoxicity prediction task, HerbToxNet achieves the best performance in all evaluation metrics, outperforming all baseline models. Compared with existing methods such as DeepDILI, QSAR-based approaches, and EI-MS-RF, HerbToxNet achieves higher AUPRC values, suggesting a better capability to identify positive samples under class-imbalanced settings. Similarly, HerbToxNet holds competitive performance in the nephrotoxicity prediction task, achieving the highest AUROC, AUPRC, and F1 scores, which demonstrates its strong discriminative ability in identifying nephrotoxic herbs. Notably, HerbToxNet obtains consistently better results across both toxicity types, validating its stable adaptability and learning capacity across different toxicological endpoints.

These results underscore that HerbToxNet is not only effective in modeling complex multi-label toxicity relationships, but also capable of extracting critical features and assessing toxicity risk in binary classification tasks.

4.4. Drug toxicity prediction

HerbToxNet is not limited to herb toxicity prediction, it can also be extended to chemical drugs. To study its generalizability, we evaluate it on the UniTox dataset (Silberg et al., 2024), which contains 1349 chemical compounds annotated with eight toxic labels. Given the structural and semantic differences between chemical drugs and herbs, we remove the heterogeneous graph and HAN, and directly use the SMILES representation of each drug as input. From the SMILES strings, we extract Morgan fingerprints to represent molecular features. These features are then integrated with the contrastive learning module and the weighted label fusion strategy to perform toxicity prediction for chemical drugs. Among the baseline methods, MulFRaL was excluded from comparison as it relies heavily on heterogeneous graph structures, which are not applicable in this setting. All other baseline methods keep unchanged. The experimental setup and evaluation metrics are the same as those used in the herb toxicity prediction task.

Table 4 summarizes the performance of HerbToxNet and baselines on predicting drug toxicity. The results of HerbToxNet are better than those of baseline methods in most evaluation metrics. It again achieves superior performance in the toxicity prediction task for chemical drugs, primarily due to its methodological strengths in both feature representation and label prediction. Although the heterogeneous graph and HAN are excluded in this setting, HerbToxNet leverages the contrastive learning module to enhance the discriminability of drug representations. Specifically, the contrastive learning strategy encourages representations of drugs with similar toxicity profiles to be closer in the feature space, this helps to distinguish different toxic labels. The use of Morgan fingerprints offers a compact and widely applicable molecular representation, which enables the model to maintain high prediction efficacy even in the absence of complex semantic contexts specific to TCM. Moreover, WLF combines the predictions from MLP and those aggregated from similar herbs, effectively alleviating the issue of limited training data. These results not only validate the generalizability of HerbToxNet, but

Table 4

Performance comparison of HerbToxNet and baseline methods in predicting drug toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.6107±0.0169	0.6961±0.0114	0.6973±0.0139	0.3610±0.0226
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
QSAR-SVM	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
EI-MS-RF	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
DeepDILI	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
R-E-GA	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
TIRPnet	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
ML-PRDF	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
MLC-CNN	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
HerbToxNet	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

also highlight the methodological adaptability of its core design across different toxicity prediction scenarios.

4.5. Ablation study and parameter sensitivity analysis

To further study the contribution factors of HerbToxNet, we introduce five variants: (i) w/o HAN directly uses the properties and efficacy data of herb as features without constructing heterogeneous graph and the meta-path, but combined with contrastive learning and weighted label fusion strategies for prediction. (ii) w/o PE randomly initializes the herb node features in the heterogeneous graph instead of using pharmacological property and efficacy data as initial features. (iii) w/o CL removes the contrastive learning module with dynamic coefficients in the training phase. (iv) w/o δ replaces the dynamic coefficient δ_{ij} with the toxicity label similarity β_{ij} . (v) w/o WLF only uses the MLP for prediction, without incorporating the weighted label fusion strategy. The details are reported in Section II of Supplementary file. Overall, outperforms its variants by a clear margin, underscoring the vital role of each removed factor in HerbToxNet for the prediction of herb toxicity.

We also conduct different experiments to study the parameter sensitivity of HerbToxNet. The results are reported and analyzed in Section III of Supplementary file.

4.6. Toxicity prediction of herbs

In this case study, we utilized HerbToxNet to predict and analyze the toxicity of canonical herbs, including *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, and *Bi Ma Zi*, in real-world scenarios. To rigorously evaluate the model's generalization capability, all other herbs in the dataset were used as the training set, while the five selected herbs were treated as a separate test set for toxicity prediction. The predicted toxicities were compared with known toxic labels, and relevant literature was retrieved from PubMed to further verify the consistency of model predictions with current research findings and traditional TCM theory.

The results demonstrate that HerbToxNet can accurately identify key toxicities of canonical hepatotoxic herbs (such as *Chai Hu* and *He Shou Wu*), effectively reveal toxicity risks in extremely “cold” or “hot” herbs (such as *Ba Dou* and *Fu Zi*), and uncover potential toxicity in herbs traditionally classified as neutral (“ping”) (such as *Bi Ma Zi*). Moreover, the toxicity predictions generated by HerbToxNet are broadly consistent with both established toxicological evidence and the classical TCM classification system, indicating that the model can leverage information encoded in traditional herb properties (nature, flavor, meridian tropism) to assess toxicity risk.

The detailed description of the experimental procedures, case prediction results, and comprehensive analyses are provided in Supplementary Section IV.

4.7. Interpretability analysis

4.7.1. Model interpretability analysis

To capture the intricate relationships between herbs, ingredients and targets, we adopt the Heterogeneous Attention Network (HAN) (Wang et al., 2019), which is particularly notable for its hierarchical attention mechanism. This mechanism enables the model to capture both the importance of a node's neighbors and the relevance of different meta-paths when learning node embeddings. To gain deeper insight into the model interpretability and to better understand the relative significance of neighbors and meta-paths, we conduct a detailed analysis of the hierarchical attention mechanism. Specifically, we take *Chai Hu* as a case study and visualize the attention scores between the node and its neighbors under different meta-paths, as illustrated in Fig. 2. From the perspective of semantic-level hierarchical attention, HAN assigns distinct attention weights to three meta-paths (HH, HH and HTH), reflecting the model's emphasis on different semantic relationships when representing the herb node *Chai Hu*. Among them, the HH meta-path receives the highest semantic attention score, indicating that herbs with shared ingredients play a dominant role in the feature learning of *Chai Hu*. In contrast, the HH meta-path receives a moderate weight, while the HTH meta-path has the lowest score. This distribution of attention scores aligns with the connectivity patterns in our dataset. The HH meta-path, with fewer but more specific associations, enables the model to capture distinctive ingredient-level features. On the other hand, the HTH meta-path, involving far more connections, leads to denser but more redundant interactions, diminishing its utility for precise herb representation. The intermediate weight for HH reflects that these herb-herb associations are indirectly inferred from pharmacological and efficacy similarities, making them less specific than HH but less redundant than HTH.

In the node-level attention mechanism, we select the top three nodes with the highest attention scores under each meta-path. For HH meta-path, *Chai Hu* is connected to *San Bai Cao*, itself, and *Feng Du*. All three herbs share the characteristic of a pungent flavor. Regarding meridian tropism, both *Chai Hu* and *Feng Du* target the liver meridian. Although *San Bai Cao* primarily targets the lung meridian, the lung and liver are physiologically interrelated, allowing for complementary therapeutic effects. Functionally, all three exhibit anti-inflammatory, anti-swelling, and analgesic properties. In addition, the connection to the self-node

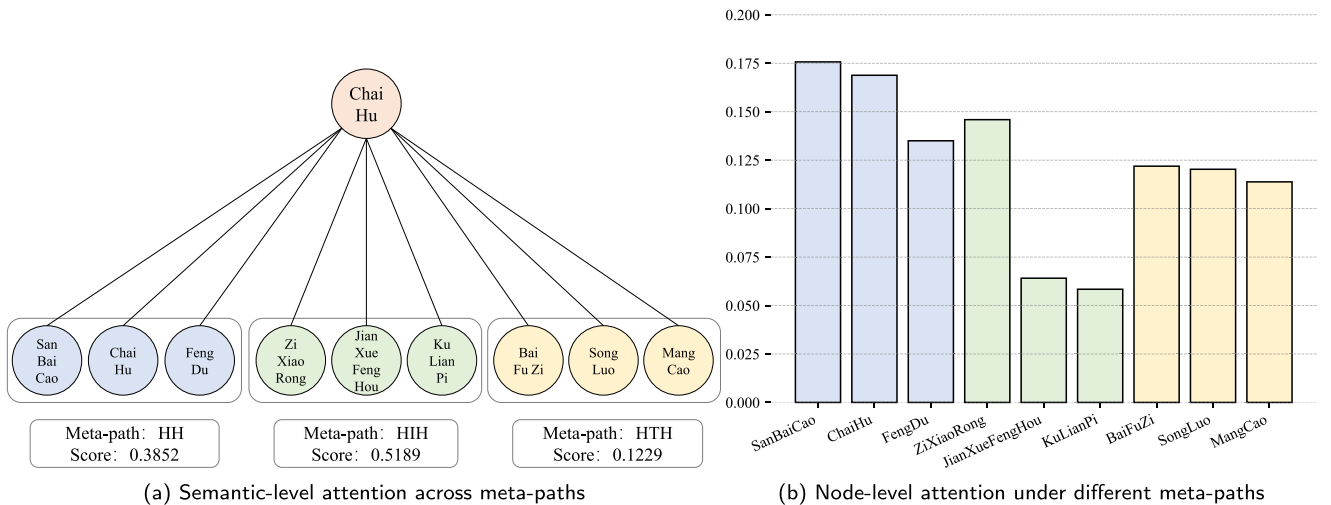


Fig. 2. Hierarchical attention analysis results of HerbToxNet on the herb node *Chai Hu*, including (a) semantic-level attention across meta-paths, and (b) node-level attention under different meta-paths.

further indicates that the node itself plays a crucial role in generating its own representation during the learning process.

For HIH meta-path, *Chai Hu* is connected to *Zi Xiao Rong*, *Jian Xue Feng Hou*, and *Ku Lian Pi*. *Chai Hu*, *Zi Xiao Rong*, *Ku Lian Pi*, and *Jian Xue Feng Hou* share common classes of secondary metabolites, which include triterpenoid saponins, flavonoids, phenolic compounds, volatile oils, and bitter alkaloids. For instance, *Chai Hu* contains saikosaponins, *Zi Xiao Rong* is rich in flavonoids and phenolic acids, *Ku Lian Pi* contains toosendanin and related limonoids, while *Jian Xue Feng Hou* harbors cardiotoxic cardiac glycosides such as antiarin. Despite their structural diversity, these ingredients share overlapped biological activities, notably anti-inflammatory, antitumor, and detoxifying effects.

For the HTH meta-path, *Chai Hu* is connected to *Bai Fu Zi*, *Song Luo*, and *Mang Cao*. These herbs share several potential pharmacological targets with *Chai Hu*, including *TNF*, *IL6*, *PTGS2*, and *MAPK1*, which are closely associated with liver injury and inflammatory responses. Saikosaponins in *Chai Hu* and diterpenoid alkaloids in *Bai Fu Zi* may upregulate pro-inflammatory cytokines such as *TNF* and *IL6* via activation of the *NFKB* signaling pathway. Usnic acid in *Song Luo* has been implicated in mitochondrial dysfunction and oxidative stress, leading to hepatotoxicity mediated by *PTGS2* overexpression. Alkaloids in *Mang Cao* may affect *MAPK1* signaling, which is involved in hepatocyte apoptosis and stress response.

In summary, the node-level attention mechanism highlights key associations based on shared herb properties, ingredients, and targets, offering biologically meaningful and interpretable insights into the model's predictions for *Chai Hu*.

4.7.2. Biointerpretability analysis

To enhance the biological interpretability of HerbToxNet, we constructed a multi-level explanatory framework along the “herb-ingredient-target-biological pathway-toxicity” axis. Taking *Chai Hu* as a representative case, we systematically identified its toxic ingredients, related molecular targets, and associated toxicity types by integrating meta-path analysis, literature curation, and toxicity databases. We further linked these components via their biological pathways and functional annotations. Our results show that HerbToxNet successfully uncovers key toxic ingredients and their molecular targets involved in hepatotoxicity, nephrotoxicity, and systemic toxicity, and reveals the relevant biological pathways and mechanisms underlying herb-induced toxic responses. This demonstrates that HerbToxNet can provide mechanistically meaningful and biologically plausible explanations for its toxicity predictions, thereby supporting more reliable risk assessment of herbs. A comprehensive description of the analytical procedures, in-depth results, and all supporting evidence can be found in Supplementary Section V.

5. Conclusion

In this paper, we construct a multi-label herb toxicity dataset and propose HerbToxNet, a framework for herb toxicity prediction that integrates heterogeneous network modeling, contrastive learning, and weighted label fusion. Empirical study shows that HerbToxNet achieves superior performance than competitive methods. HerbToxNet can identify novel potential toxicities and improve the safe clinical use of herbs. Moreover, it can uncover toxicity-related ingredients and targets, provide insights into toxicity mechanisms, guide antidote development, support toxicity-related databases construction and the discovery of safer herbs. Although HerbToxNet achieves better performance than compared methods and gives some authentic results, it still has limitations. Our dataset currently covers five toxicity categories, with many herbs assigned multiple labels, supporting the use of multi-label classification. Due to limited standardized data, broader toxicity categories are not included at this stage. In future work, we will expand toxicity coverage to enhance the robustness and applicability of HerbToxNet.

CRedit authorship contribution statement

Yongzheng Zhu: Data curation, Formal analysis, Investigation, Software, Validation, Writing – original draft; **Yunbo Miao:** Formal analysis, Software, Validation, Writing – original draft; **Rong Sun:** Supervision, Resources; **Zhongmin Yan:** Methodology, Formal analysis, Writing – review & editing; **Guoxian Yu:** Supervision, Investigation, Methodology, Formal analysis, Writing – review & editing.

Data Availability

Our code and collected dataset are shared at <https://www.sdu-idea.cn/codes.php?name=HerbToxNet>.

Declaration of competing interest

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.eswa.2025.129969](https://doi.org/10.1016/j.eswa.2025.129969).

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